TASK-1 : Run these Programs and observe their output.

```
#include<iostream>
using namespace std;
int main ( )
{
  Char letter;
  letter = 'A';
  cout << letter << '\n';
  letter = 'B';
  cout << letter << '\n';
  cin >> letter;
  return 0;
}
```

Output :

A

B

```
#include <iostream>
using namespace std;
int main ( )
{
  char letter;
  letter = 65;
  cout << letter << '\n';
  letter = 66;
  cout << letter << '\n';
  cin >> letter ;
  return 0;
}
```

Output :

A

B

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```
#include <iostream>
using namespace std;
int main ()
{
  int letter;
  letter = 'A';
  cout << letter << '\n';
  letter = 'B';
  cout << letter << '\n';
  cin >> letter;
  return 0;
}
```

Output:

```
65
66
```

TASK-2:

Take two integers as input from the user apply arithemadic Operations on them (+,-,*,/) as print them on Screen.

```
#include <iostream>
using namespace std;
int main ()
{
  int num1, num2;
  cout << "Enter first integer:";
  cin >> num1;
  Cout << "Enter Second integer:";
  cin >> num2;
```

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```
cout << " Addition = " << num1 + num2 << endl;
cout << " Subtraction = " << num1 - num2 << endl;
12. cout << " Multiplication = " << num1 * num2 << endl;
13. if (num2 != 0)
14. cout << " Division = " << num1 / num2 << endl;
15. else
16. cout << " Division is not possible by Zero. " << endl;
17. return 0;
18. }
```

Output :

```
Enter First integer : 10
Enter Second integer : 2
Addition = 12
Subtraction = 8
Multiplication = 20
Division = 5
```

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Task-3:

Write the output of the Following code.

```
1. #include "stdafx.h"
2. #include <iostream>
3. using namespace std;
4. int main ()
5. {
6.   int val1;
7.   int val2;
8.   int val3;
9.   coul << "Please enter your 3 numbers:";
10.  cin >> val1 >> val2 >> val3;
11.  cout << val1 << "\n" << val2 << "\n" << val3 << "\n";
12.  cout << val3 << "\n" << val2 << "\n" << val1 << "\n";
13.  System ("PAUSE");
14.  return 0;
15. }
```

Output

```
Please enter your 3 numbers: 1 2 3
1
2
3
3
2
1
```

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Task-4:

Write a C++ Program to calculate the distance between two points.
Note: x1, y1, x2, y2 are all double values.

Formula:

$$d = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

```
1. #include <iostream>
2. H include <cmath>
3. using namespace std;
4. int main () {
5.    double x1, Y1, X2, Y2;
6.    cout << "Enter X1 Y1 X2 Y2 :";
7.  cin >> x1 >> Y1 >> X2 >> Y2 ;
8.  double d = sqrt (Pow (X2 - X1, 2) + Pow (Y2 - Y1, 2));
9.  cout << " Distance = " << d;
10.   return 0;
11.  }
```

Output:

```
Enter X1 Y1 X2 Y2 : 0 0 3 4

Distance = 5
```

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Task: 5

1. Calculate the Area of a Circle (area = PI * r2)

```
1. #include <iostream>
2. using nameSpace std;
3. int main () {
4.    double r= 5;
5.    double PI = 3.14159;
6.    double area = PI * r * r;
7.    cout << "Area of Circle =" << area;
8.    return 0;
9. }
```

Output:

Area of Circle = 78.5397

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Calculate the area of a Rectangle (area = length * width)

```
1. #include <iostream>
2. using namespace std;
3. int main ( )
4. {
5.   double length = 8;
6.   double width = 5;
7.   double area = length * width;
8.   cout << "length = " << length << endl;
9.   cout << " width = " << width << endl;
10.  cout << "Area of Rectangle = " << area << endl;
11.  return 0;
12. }
```

Out put :

Length = 8

width = 5

Area of Rectangle = 40

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Calculate the Area of Triangle (area = base * height * 0.5)

```
1. #include <iostream>
2. using namespace std;
3. int main ()
4. {
5.    double base = 4;
6.    double height = 3;
7.    double area = base * height * 0.5;
8.    cout << " Area of Triangle = " << area;
9.    return 0;
10. }
```

Output :

Area of Triangle = 6

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ASK-6: Write a program that holds the Prices of Five items in Five variables.....

```
#include <iostream>
#include <iomanip>
using namespace std;
int main ( ) {
    double item1 = 12.95;
    double item2 = 24.95;
    double item3 = 6.95;
    double item4 = 14.95;
    double item5 = 3.95;
    double Subtotal = item1 + item2 + item3 + item4 + item5;
    double SalesTax = Subtotal * 0.06;
    double total = Subtotal + SalesTax;
    cout << fixed << setPrecision(2);
    cout << "Price of item1 = $" << item1 << endl;
    cout << " Price of item2 = $" << item2 << endl;
cout << "Price of item 3 = $" << item3 << endl;
cout << "Price of item4 = $" << item4 << endl;
cout << "Price of item 5 = $" << item5 << endl;
cout << "Subtotal = $" << Subtotal << endl;
cout << " SalesTax (6%) = $" << SalesTax << endl;
```

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```
cout << "Total = $" << total << endl;
return 0;
}
```

Out Put :

Price of item1 = $12.95
Price of item2 = $ 24.95
Price of item 3 = $ 6.95
Price of item4 = $ 14.95
Price of item 5 = $ 3.95
Subtotal = $ 63.75
Sales Tax (6%) = $ 3.82
Total = $ 67.58

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BONUS TASK: Write a program that inputs a five-digit integer, separates the integer into its individual dig and prints the digits separated from one another by three spaces each.

```
1. #include <iostream>
2. using namespace std;
3. int main() {
4. int num;
5. cout << "Enter a five-digit integer: ";
6. cin >> num;
7. cout << num / 10000 <<  "   ";
8- Cout << (num / 1000) % 10 <<  "   ";
9- cout << (num / 100) % 10 <<  "   ";
10. cout << (num / 10) % 10 <<  "   ";
11. cout << num % 10 << endl;
12.  return 0;
13. }
```

Output:

```
Enter a five-digit integer: 12345
1   2   3   4   5
```

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ome Tasks:

Task-1: Exchange write a program that reads in two va
and swaps the value of these variables....

```
1. #include <iostream>
2. Using nameSpace std;
3. int main () {
4.     int var1, var2, temp;
5.  cout << " Enter value of var1:";
6.  cin >> var1;
7.  cout << " Enter value of var2:";
8.  cin >> var2;
9.  cout << "\nBefore swapping:" << endl;
10.  cout << " var1=" << var1 << endl;
11.  cout << " var2=" << var2 << endl;
12.      // swapping
13.  temp = var1;
14.  var1 = var2;
15.  var2 = temp;
16.  cout << "\nAfter swapping:" << endl;
17.  cout << "var1=" << var1 << endl;
18.  cout << "var2=" << var2 << endl;
```

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```
    return 0;
}
```

Out Put:

Enter value of var1 : 10

Enter value of var2 : 20

Before swapping :

var1 = 10

var2 = 20

After swapping :

var1 = 20

var2 = 10

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SK-2: Make Upper Write a program that prompts the use
enter an alphabet in small caps (a, b, c, ... z
display the entered alphabet into its upper case. (A,

```
1. #include <iostream>
2. Using namespace std;
3. int main() {
4.     char ch;
5.     cout << "Enter a small alphabet : ";
6.       cin >> ch;
7.     ch = ch - 32;
8.   cout << "Upper case alphabet : " << ch << endl;
9.   return 0;
10.    }
```

Out put :

Enter a small alphabet : b

Uppercase alphabet : B

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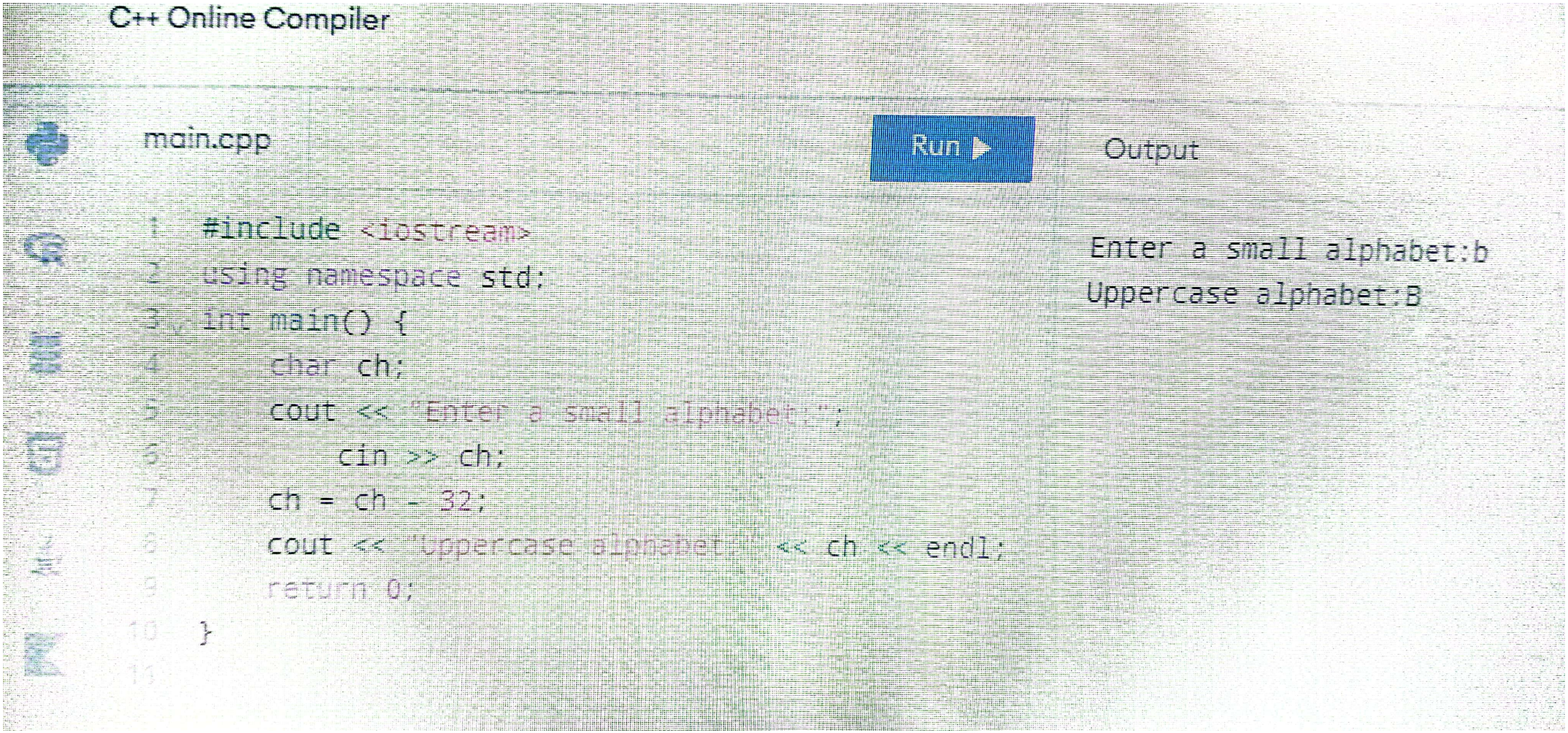

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main.cpp

Run ▶

```cpp
#include <iostream>
using namespace std;
int main() {
    int var1, var2, temp;
    cout << "Enter value of var1:";
    cin >> var1;
    cout << "Enter value of var2:";
    cin >> var2;
    cout << "\nBefore swapping:" << endl;
    cout << "var1 =" << var1 << endl;
    cout << "var2 =" << var2 << endl;
    // swapping
    temp = var1;
    var1 = var2;
    var2 = temp;
    cout << "\nAfter swapping:" << endl;
    cout << "var1 = " << var1 << endl;
    cout << "var2 = " << var2 << endl;
```

Output

```
Enter value of var1:10
Enter value of var2:20

Before swapping:
var1 =10
var2 =20

After swapping:
var1 = 20
var2 = 10
```

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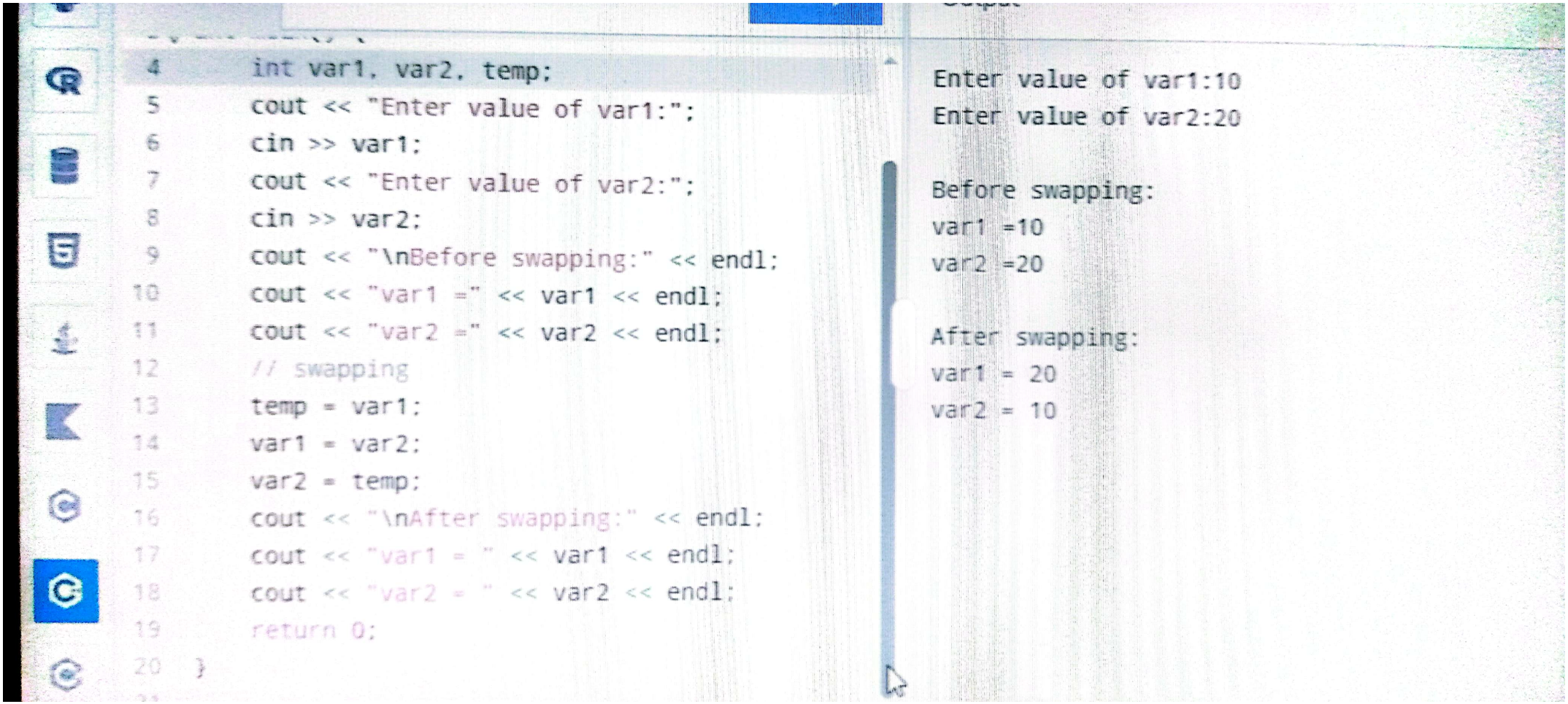

```
4       int var1, var2, temp;
5       cout << "Enter value of var1:";
6       cin >> var1;
7       cout << "Enter value of var2:";
8       cin >> var2;
9       cout << "\nBefore swapping:" << endl;
10      cout << "var1 =" << var1 << endl;
11      cout << "var2 =" << var2 << endl;
12      // swapping
13      temp = var1;
14      var1 = var2;
15      var2 = temp;
16      cout << "\nAfter swapping:" << endl;
17      cout << "var1 = " << var1 << endl;
18      cout << "var2 = " << var2 << endl;
19      return 0;
20  }
```

```
Enter value of var1:10
Enter value of var2:20

Before swapping:
var1 =10
var2 =20

After swapping:
var1 = 20
var2 = 10
```

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main.cpp

Run ▶

```cpp
#include <iostream>
using namespace std;
int main() {
    int num;
    cout << "Enter a five-digit integer:";
    cin >> num;
    cout << num / 10000 << "   ";
    cout << (num / 1000) % 10 << "   ";
    cout << (num / 100) % 10 << "   ";
    cout << (num / 10) % 10 << "   ";
    cout << num % 10 << endl;
    return 0;
}
```

Output

```
Enter a five-digit integer:12345
1   2   3   4   5
```

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C++ Online Compiler

main.cpp

Run

```cpp
#include <iostream>
#include <iomanip>
using namespace std;
int main() {
    double item1 = 12.95;
    double item2 = 24.95;
    double item3 = 6.95;
    double item4 = 14.95;
    double item5 = 3.95;
    double subtotal = item1 + item2 + item3 + item4 + item5;
    double salesTax = subtotal * 0.06;
    double total = subtotal + salesTax;
    cout << fixed << setprecision(2);
    cout << "Price of item1 = $" << item1 << endl;
    cout << "Price of item2 = $" << item2 << endl;
    cout << "Price of item3 = $" << item3 << endl;
    cout << "Price of item4 = $" << item4 << endl;
    cout << "Price of item5 = $" << item5 << endl;
```

Output

```
Price of item1 = $12.95
Price of item2 = $24.95
Price of item3 = $6.95
Price of item4 = $14.95
Price of item5 = $3.95
Subtotal = $63.75
SalesTax(6%) = $3.82
Total = $67.58
```

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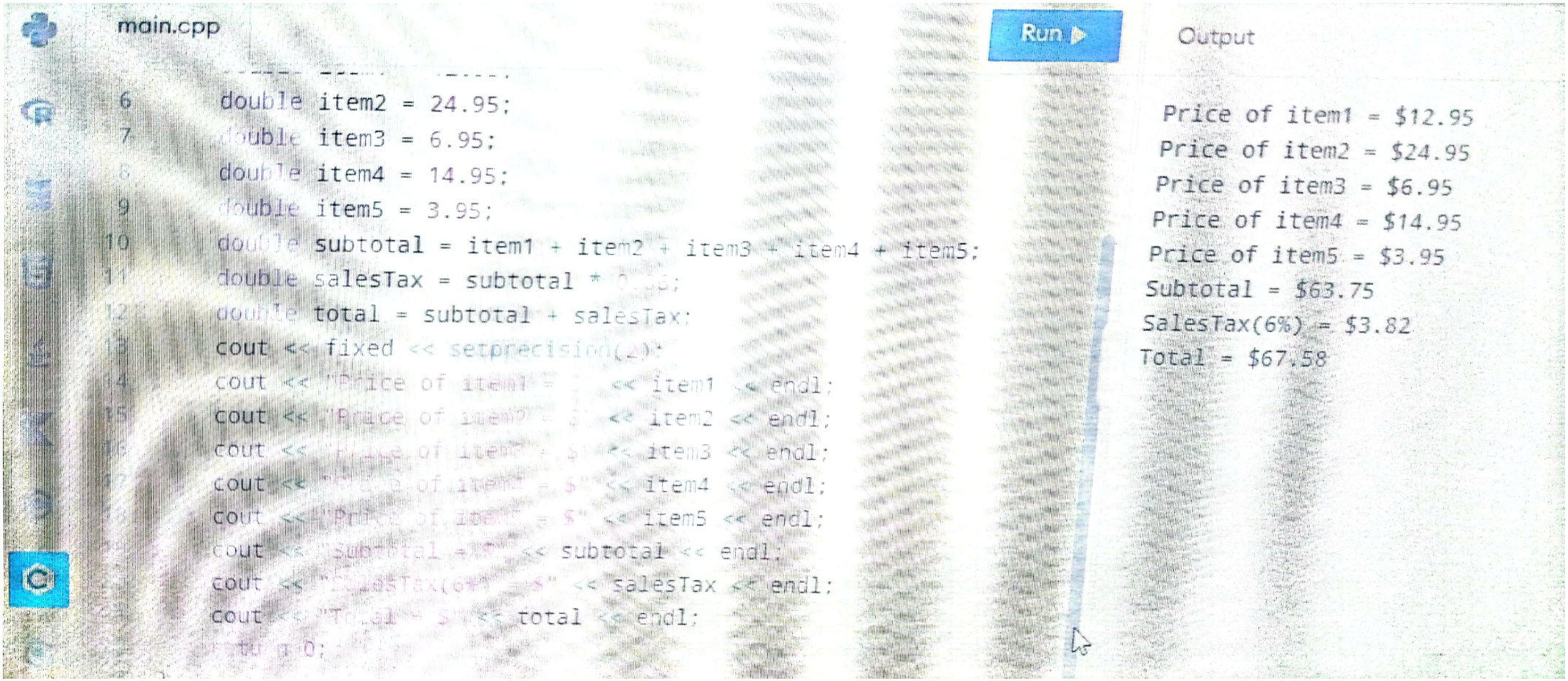

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C++ Online Compiler

main.cpp

```cpp
#include <iostream>
using namespace std;
int main() {
    double length = 8;
    double width = 5;
    double area = length * width;
    cout << "Length =" << length << endl;
    cout << "Width =" << width << endl;
    cout << "Area of Rectangle =" << area << endl;
    return 0;
}
```

Run

Output

```
Length =8
Width =5
Area of Rectangle =40
```

Scanned with CamScanner

main.cpp

Run

Output

```cpp
#include <iostream>
using namespace std;
int main () {
    int val1 = 10;
    int val2 = 20;
    int val3 = 30;
    cout << "10 20 30:";
    cout << val1 << "\n" << val2 << "\n" << val3 << "\n";
    cout << val3 << "\n" << val2 << "\n" << val1 << "\n";
    return 0;
}
```

```
10 20 30:10
20
30
30
20
10
```

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main.cpp

Run

Output

```cpp
#include <iostream>
#include <cmath>
using namespace std;
int main () {
    double x1, y1, x2, y2;
    cout << "Enter x1 y1 x2 y2:";
    cin >> x1 >> y1 >> x2 >> y2;
    double d = sqrt(pow(x2 - x1, 2) + pow(y2 - y1, 2));
    cout << " Distance =" << d;
    return 0;
}
```

```
Enter x1 y1 x2 y2:0 0 3 4
 Distance =5
```

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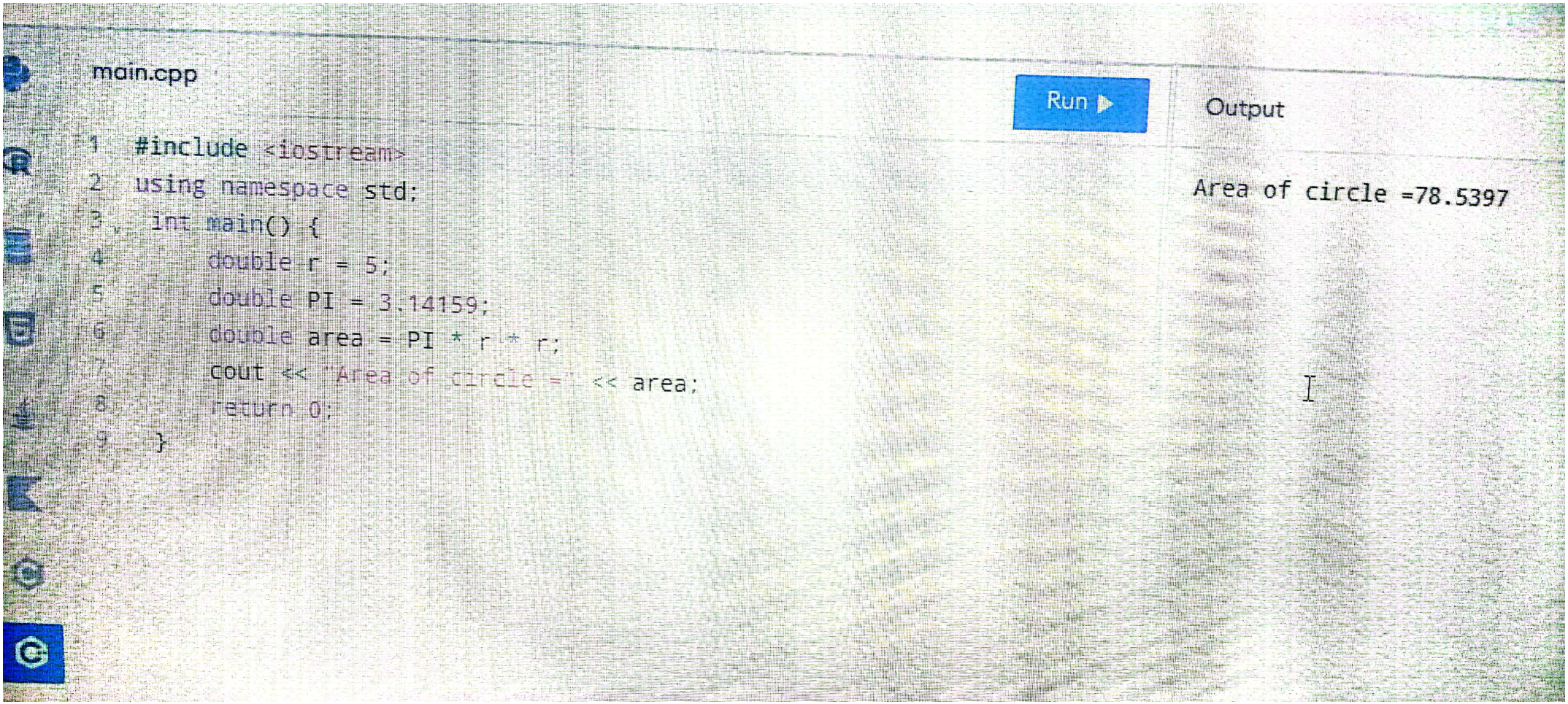

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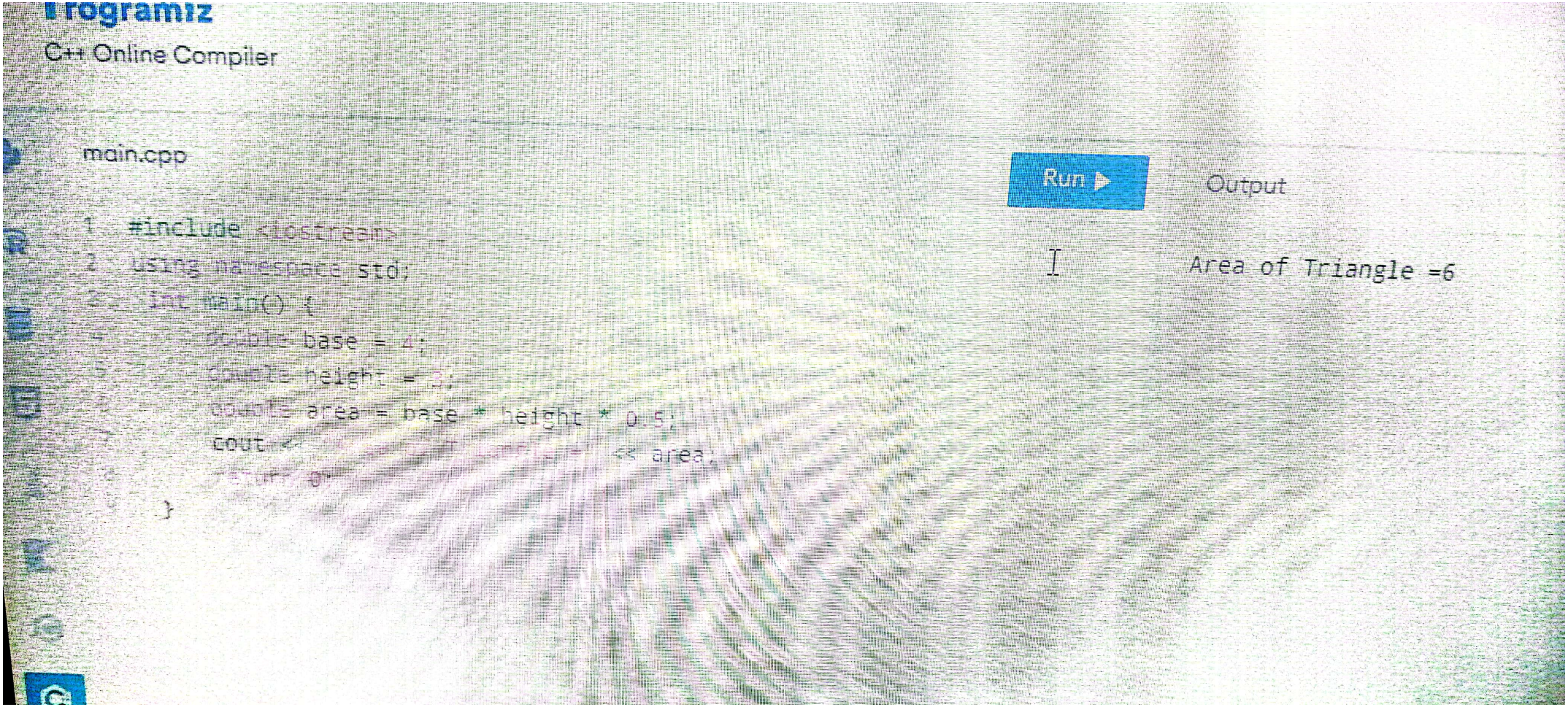

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C++ Online Compiler

Programiz

main.cpp

Run

Output

```
#include <iostream>
using namespace std;
 int main() {
      double r = 5;
      double PI = 3.14159;
      double area = PI * r * r;
      cout << "Area of circle =" << area;
      return 0;
  }
```

```
/tmp/NRKFQHsUG9-KHN5NKTt/main.cpp:1:2: error: invalid preprocessing
directive #iclude; did you mean #include?
    1 | #iclude <iostream>
      |  ^~~~~~
      |  include

/tmp/NRKFQHsUG9-KHN5NKTt/main.cpp: In function 'int main()':
/tmp/NRKFQHsUG9-KHN5NKTt/main.cpp:7:6: error: 'cout' was not declared
in this scope
    7 |      cout << "Area of circle =" << area;
      |      ^~~~

+++ |+#include <iostream>
  1 | #iclude <iostream>
```

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C++ Online Compiler

main.cpp

Run ▶

Output

```
#include <iostream>
#include <cmath>
using namespace std;
int main () {
    double x1, y1, x2, y2;
    cout << "Enter x1 y1 x2 y2:";
    cin >> x1 >> y1 >> x2 >> y2;
    double d = sqrt(pow(x2 - x1, 2) + pow(y2 - y1, 2));
    cout << " Distance =" << d;
    return 0;
}
```

```
Enter x1 y1 x2 y2:
```

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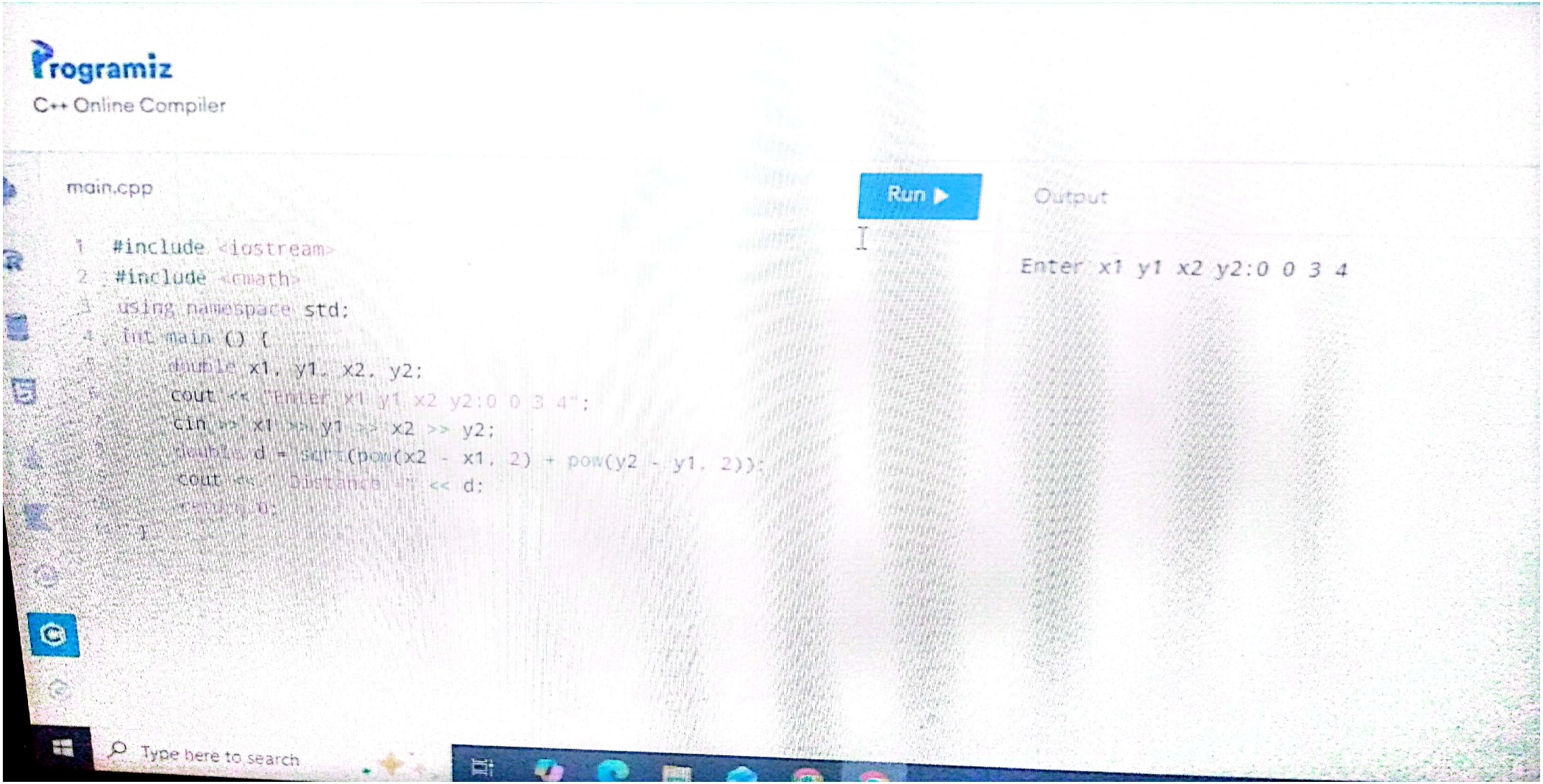

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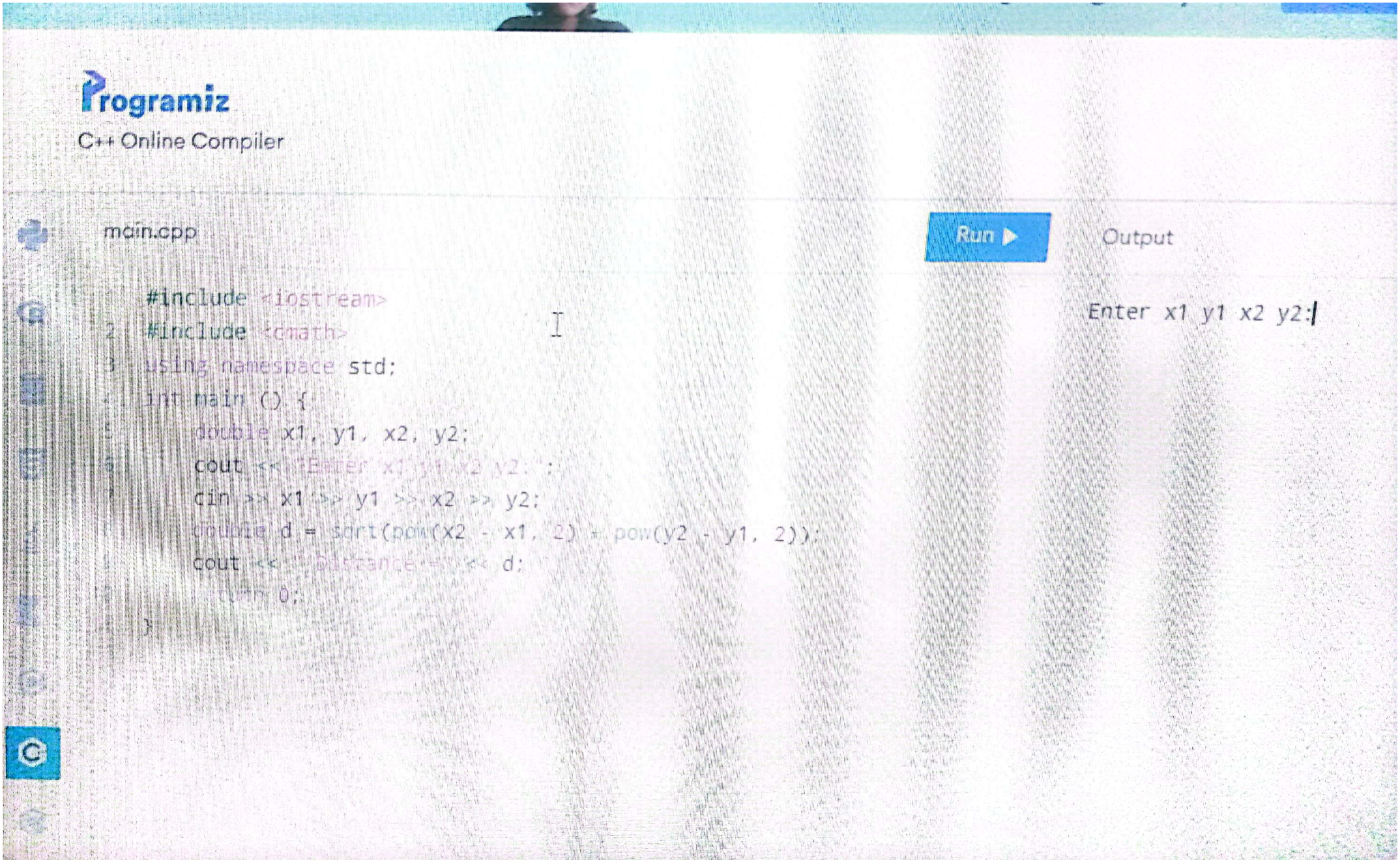

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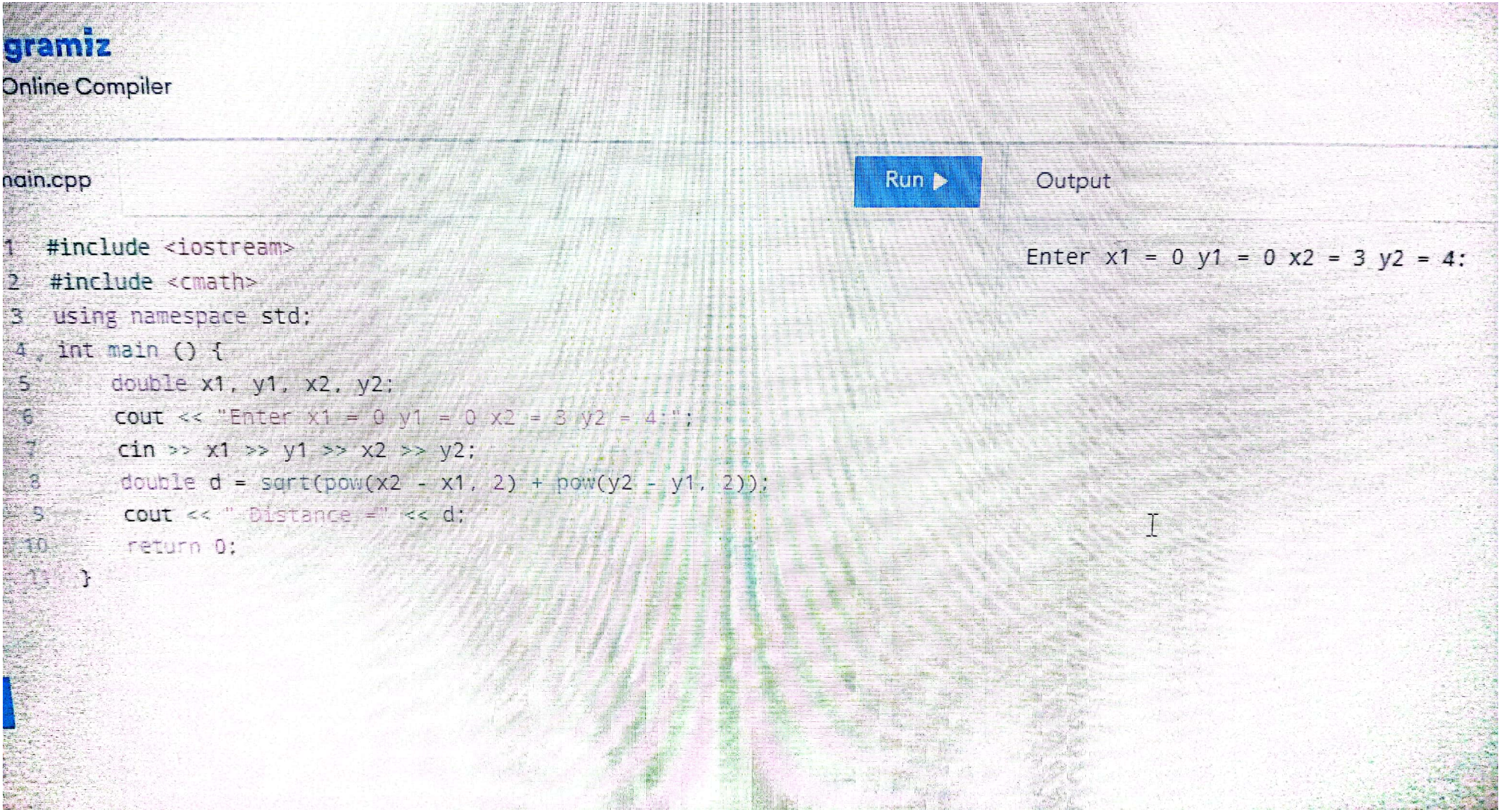

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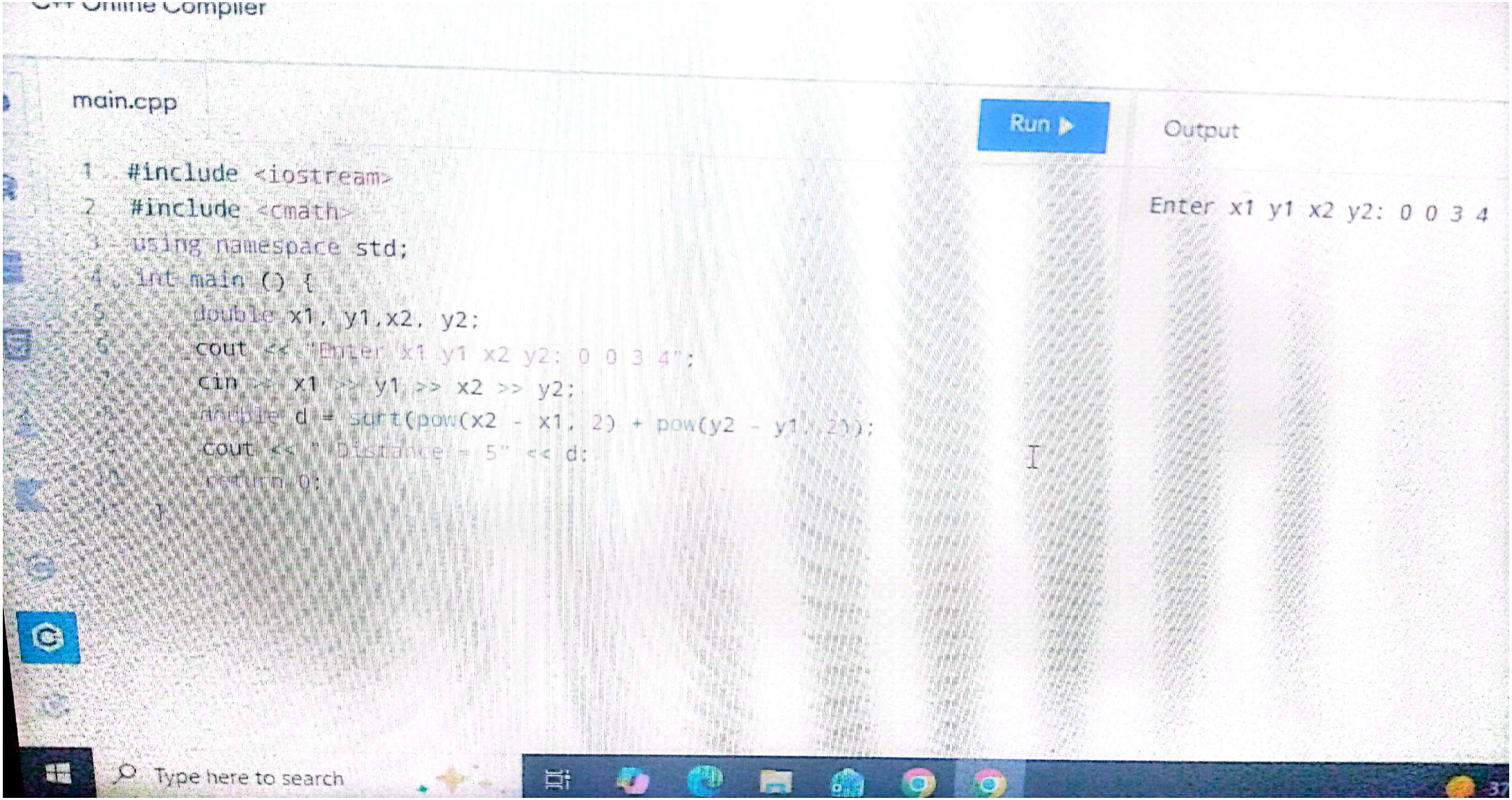

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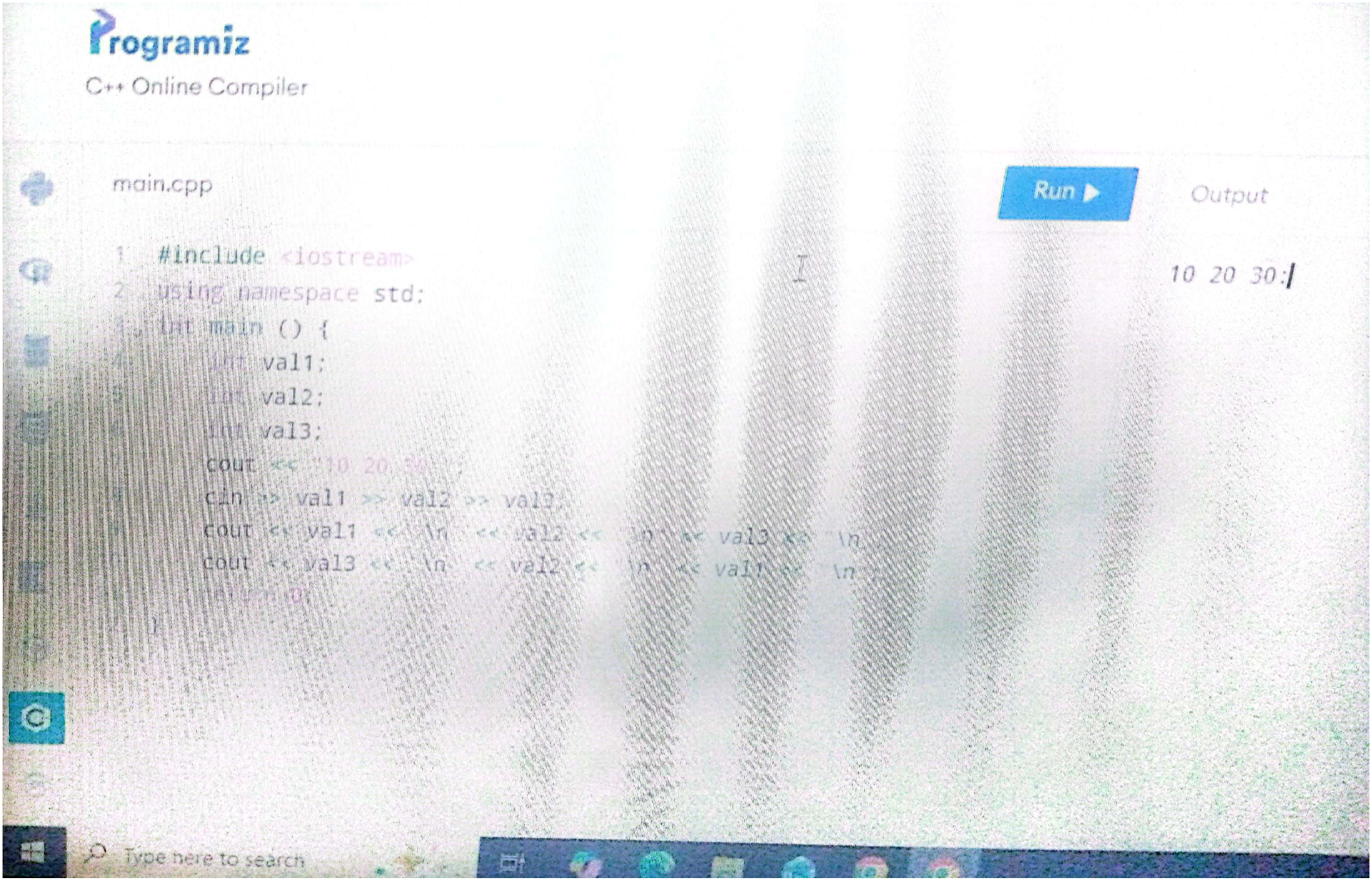

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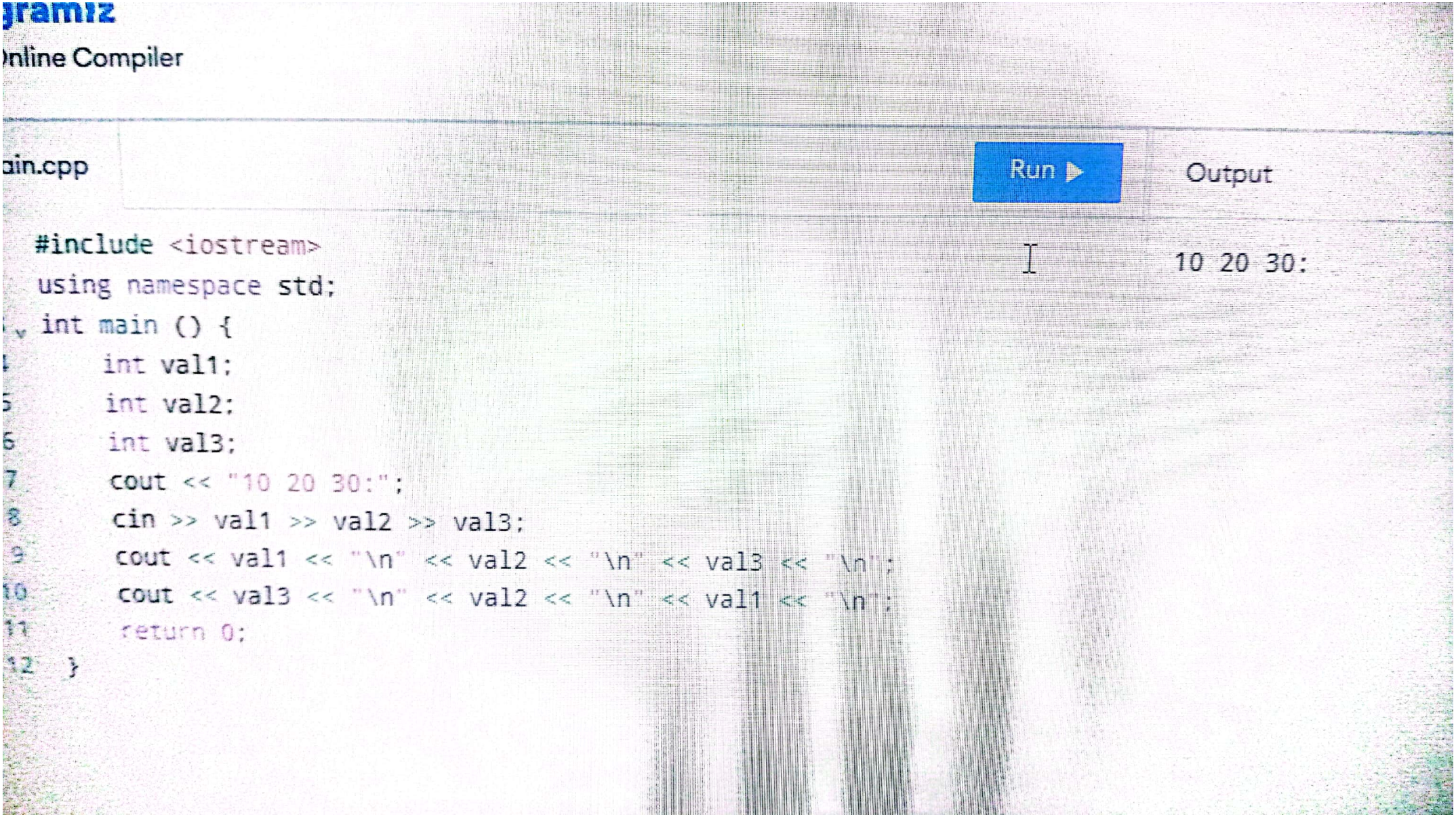

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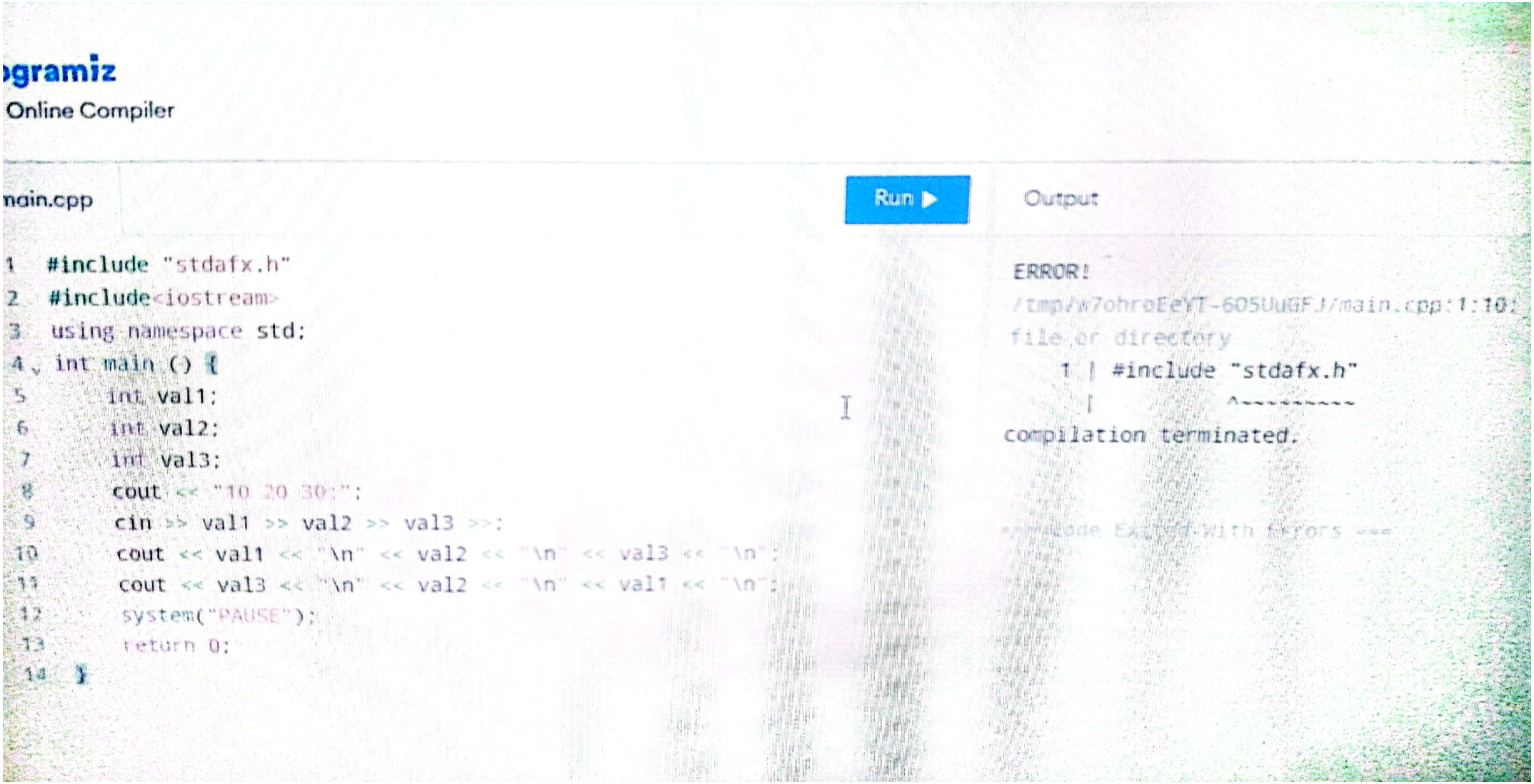

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